Randall Preston

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EDUCATION

Villanova University | College of Engineering | Villanova, PA **May 2029**
Bachelor of Science, Mechanical Engineering | *Business Minor*
Relevant Coursework: Engineering CAD/Programming, Statics, Circuits, Applied Linear Algebra, Diff. Equations, Calculus
GPA: 3.88/4.0 | Dean's List (Fall 2025-Present)

Summer Business Institute | Villanova University – Business Minor **Summer 2026**
16-Credit, 10-week intensive program focused on key concepts in accounting, business law, finance, economics, analytics, information technology, management, marketing, and professional development

ENGINEERING EXPERIENCE

NovaRacing FSAE | *Drivetrain System Design Engineer* | Villanova University **October 2025-Present**

VU18 Design Process **Spring 2026-Present**

- Leading iterative design of VU18 drivetrain, balancing mass, cost, and manufacturability within FSAE rules
- Implementing agentic AI into ANSYS FEA/topology optimization workflow for differential-engine mount design, enabling remote simulation setup, increased simulation capability, and faster design validation
- Upgrading half-shafts from steel to composite design for a projected 58% mass reduction
- Upgrading diff-mount turnbuckles from steel to titanium/aluminum for potential 25-50% mass reduction

VU17 Design Process **Winter-Spring 2026**

- Designed, manufactured, and assembled drivetrain components, supported turnbuckle tensile testing to ~14 kN
- Assisted in manufacturing 30+ carbon-fiber aerodynamic components for VU17 and a Senior Capstone Project, from CNC-routed MDF molds through wet layup/resin infusion, vacuum bagging, demolding, and assembly

Undergraduate Research Scholar Program | Villanova University **March 2026-Present**

Hydrogen Permeation Testing in Structural Metals

- Conducts research with faculty advisor and team of six undergraduate and doctoral researchers on hydrogen permeation of structural metals for automotive and energy industry applications
- Designs and validates Devanathan-Stachurski dual-cell electrochemical permeation test setup, including hardware specifications and solution chemistry for hydrogen generation and permeation detection
- Collaborates on sample creation process and SEM imaging for material failure analysis/characterization, using result discussion to inform in-situ experimental setup and future permeation testing parameters
- Meets weekly with team and advisor to share updates, present findings, and coordinate lab work

LEADERSHIP & ACTIVITIES

Pi Kappa Alpha Fraternity | *Historian & Real Estate Chair* | Villanova University **April 2026-Present**

- Oversees meeting/event documentation, year-end report, and Petition to Install for provisional chapter
- Organizes biweekly meetings with six-member real estate committee and alumni advisors

Villanova Band | *Trumpet Player* | Villanova University **September 2025-Present**

- Attends weekly rehearsals, plays at athletic and admissions events

Scouting BSA | *Eagle Scout* | Troop 53, Alton, NH **2019-2026**

SKILLS

Design & Analysis | CAD (SolidWorks); MATLAB; ANSYS (Structural, CFD - Novice); Excel
Manufacturing | Carbon Fiber Composites (Wet Layup, Resin Infusion, Post-Processing, Assembly); GD&T; Machining (Manual Mill, Manual Lathe); Precision Measurement; Soldering; Welding – Novice
Materials & Testing | Electrochemical Testing (Potentiostat, Galvanostat); Hydrogen Permeation Testing (Devanathan-Stachurski, ASTM G148); Tensile Testing; Test Setup Design & Validation; Instrument Troubleshooting